

T-OCS V6.0

TIME-PROGRAMMABLE OCCUPANT CONTROL SEAT

Patent Preparation Draft & Invention Disclosure - V6.0

Engineering focus: time-dependent state transitions + decoupled ride-down and seatback-rotation control + redundant bilateral load paths + explicit rebound management + occupant trajectory management.

Inventor: [TO BE SUPPLIED] **Applicant/Assignee:** [TO BE SUPPLIED]
Priority/Filing data: [TO BE SUPPLIED] **Docket:** T-OCS-V6-PAT-001
Version: V6.0 | 2026-09-02

Ruthless status: V6 is materially stronger than V5 as a patent-preparation package because it removes weak "artificial gravity" framing from the core claims, narrows the inventive center around specific mechanical relationships, and maps each claimed feature to a physical/CAE observable. It still is not a patent grant, not a legal opinion, and not a validated safety system.

Key rule: never present a target, illustrative curve, or generated CAD visualization as a measured crash result.

1. Executive Summary

The V6 disclosure treats the invention as a vehicle-seat architecture that controls occupant trajectory and the time-history of force/energy transfer. The system uses a mechanically distinct ride-down path for controlled seat translation and a separate rotation-control path for the seatback. A multi-state mechanism changes the seat between normal, armed/capture, ride-down, rebound-control and secure states. The intended benefit is not creation of a new gravitational field; the effective-gravity language is retained only as an inertial-frame interpretation.

V6 inventive center: a coordinated combination of state transition, seat translation, independent seatback rotation control, redundant bilateral load paths, and explicit rebound control.

The original project materials support this engineering direction. The V5 design report defines a dual-rail carriage, staged compliance, rotation control, pelvic anti-submarining geometry, and dedicated rebound control, while explicitly classifying the geometry as a research prototype. ■filecite■turn9file0■L681-L681■

The supplied 2025 NHTSA/SAE rear-impact study is especially relevant: it reported collapse-like kinematics at a 25.6 g scaled pulse and a reduction in seatback rotation from 47.5 degrees to 24 degrees in a strengthened model, with reduced ramping and lower modeled HIC/Nij in that simulation. These values belong to that specific research model and are not claimed as T-OCS performance guarantees. ■filecite■turn5file5■L526-L530■ ■filecite■turn5file5■L542-L552■

The THOR-05F material is used only to learn validation discipline. It documents extensive instrumentation, repeated qualification, explicit seating procedures, and post-test inspection/requalification; those methods inform the validation plan but are not claimed as T-OCS technology. ■filecite■turn2file4■L286-L299■

2. Ruthless Patentability Assessment - What V6 Must Avoid

The V5 claim direction was still too vulnerable if interpreted broadly as "vehicle seat + rail + absorber + inertial lock." The current Google Patents landscape contains direct disclosures covering important blocks of that combination.

Feature	Prior-art pressure	V6 treatment
Rail-based longitudinal energy attenuation	Very high	Not claimed alone; tied to time-dependent state logic and a distinct rotation-control path.
Rotary/track energy attenuation	Very high	Not claimed as the inventive center.
Crash-triggered rail release + absorber	Very high	Not claimed as the sole novelty; used only as a dependent/alternative implementation.
Inertial pawl/cam lock	High	Not claimed generically; tied to the T-OCS multi-state architecture.
Anti-submarining geometry	High	Dependent feature only; linked to bilateral load-path and occupant-trajectory control.
Front/rear modular packaging	Low novelty by itself	Embodiment/application layer, not core novelty.
Decoupled translation-energy path	Potentially differentiating	Core independent-claim limitation.
Explicit rebound-control state affected by differentiation	Potentially differentiating in combination	Core limitation / dependent claim family.
State transition tied to dynamic measurement at variable rates	Potentially differentiating	Core limitation, with physical observables.

Prior-art anchors verified from Google Patents: US20200231073A1 discloses track-based energy attenuation with independently pivotable seat pan/back and rotary attenuation; WO2021197809A1 discloses a rail pair, crash-triggered release, energy absorber and locking device; US4634182A covers inertia-operated recliner locking; US6050635A covers anti-submarining seat-pan structure.
■cite■turn992619search0■turn464299search0■turn992619search1■turn992619search2■

Critical conclusion: the claim should not say or imply that T-OCS is novel merely because it uses rails, a damper, an inertial trigger, a cam/pawl, a seatback hinge, or anti-submarining features. Those are crowded elements.

3. V6 Claim Architecture - Narrower and Harder to Attack

The V6 claim family is organized around relationships that must occur together. The intended protection is not a parts shopping list; it is the architecture and sequence by which the parts interact.

Independent apparatus claim - proposed center:

1. A vehicle seat system comprising a seat pan and a seatback coupled to a vehicle structure through at least two laterally separated longitudinal load paths; a first energy-management path configured to permit controlled longitudinal displacement of a seat carriage relative to the vehicle structure during a dynamic event; a second mechanically distinct seatback-rotation control path configured to limit seatback rotation independently of the controlled longitudinal displacement; and a state-transition mechanism configured to place the seat system, in response to a detected or predetermined dynamic event, through at least an armed state, a ride-down state, and a rebound-control state, wherein transition from the ride-down state to the rebound-control state is associated with a defined travel or kinematic condition and the occupant restraint path remains operable independently of the first energy-management path.

Why this is stronger: it forces the examiner to confront the *combination* of (i) bilateral load paths, (ii) controlled translation, (iii) a separate rotation-control path, and (iv) an explicit rebound transition tied to an actual kinematic condition. It does not depend on the phrase "effective gravity."

Independent method claim - proposed center:

16. A method of managing occupant motion in a vehicle seat comprising maintaining a secured operating state during ordinary use; detecting or determining a dynamic event; transitioning to an armed or capture state; permitting controlled seat translation through a characterized energy-management path; independently limiting seatback rotation through a distinct mechanical path; changing to a rebound-control state after a predetermined travel or kinematic condition; and maintaining an occupant restraint path during the translation and rebound-control states.

Independent cartridge claim - proposed center:

21. A serviceable seat energy-management cartridge comprising a guided moving member, a progressive resistance assembly, a damping valve assembly, a first seat-side interface and a second vehicle-side interface, the cartridge being configured for placement in a longitudinal seat load path and characterized by a force-stroke relationship having at least a first lower-resistance region, a second energy-absorption region, and a third end-control region, wherein a separate seatback-rotation control path can operate without requiring the cartridge to become the structural seatback hinge.

4. Proposed Dependent Claims - V6 Candidate Set

2. The system of claim 1, wherein the first energy-management path is provided on each of the left and right longitudinal load paths.
3. The system of claim 1, wherein the second seatback-rotation control path comprises bilateral links, dampers, torsion elements, hard stops, a locking interface, or a combination thereof.
4. The system of claim 1, wherein the state-transition mechanism includes a normal-driving state before event recognition.
5. The system of claim 1, wherein the ride-down state permits a bounded seat stroke while the restraint path remains structurally available.
6. The system of claim 1, wherein the rebound-control state applies directional damping, friction, mechanical locking, or a combination thereof against reverse seat motion.
7. The system of claim 1, wherein the transition to ride-down depends on a measured or mechanically inferred seat acceleration, vehicle acceleration, seat travel, seatback rotation, or combination thereof.
8. The system of claim 1, wherein a pelvic control surface is configured to resist forward pelvic migration while distributing contact load across a seat-pan region.
9. The system of claim 1, wherein torso guidance surfaces are compliant and configured to guide motion without constituting a concentrated rigid impact surface.
10. The system of claim 1, wherein the seatback rotation-control path has a limiting function that is mechanically independent from the force-stroke response of the longitudinal energy-management path.
11. The system of claim 1, wherein the two longitudinal load paths are independently instrumentable or independently modelable for comparison of reaction loads.
12. The system of claim 1, wherein the energy-management path comprises a replaceable cartridge.
13. The system of claim 12, wherein the replaceable cartridge comprises hydraulic, pneumatic, viscoelastic, frictional, metallic-crush, composite-crush or hybrid energy dissipation.
14. The system of claim 1, wherein the seat is configurable as a front-row seat or a rear-row seat without changing the state-transition logic.
15. The system of claim 14, wherein the rear-row seat is implemented as an independently moving 2+1 module or as three independently managed seat modules.

5. V6 Additional Development Package - Engineering Additions

Addition	Engineering change	Purpose / problem addressed	Claim relevance
Temporal state machine	Explicit S0/S1/S2/S3/S4/S5 states with transitions	Turns "time" from a slogan into a controllable mechanical state	Core state variable.
Decoupled rotation path	Separate rotation-control linkage from ride-down absorber	Avoids the false choice between making the whole seat rotate and allowing collapse/ramp in	Core
Dual bilateral load paths	Left/right rails + independent reaction routes	Redundancy, torsional stability, measurable load sharing	Core/dependent.
Rebound-control phase	Directional damper/lock after primary stroke	Reduces secondary excursion and interior-contact risk	Core/dependent.
Serviceable absorber cartridge	Replaceable characterized module	Enables repeatable component testing and post-event replacement	Quick change claim family.
Pelvis control geometry	Distributed anti-submarining bowl + load-guiding	Consolidates forward pelvic migration without a hard point contact	Dependent.
Instrumentation interfaces	Seat travel, rotation, hardpoint, belt and reaction load	Creates and discretizes location limitations to validation	Disclosure/enablement support.
Front/rear modularity	Common logic with different packaging modules	Supports commercial scalability without making seat count an inventive feature.	Enabling

The V6 engineering basis matches the source design package: dual-rail ride-down, progressive absorption, separate rotation control, pelvic anti-submarining, rebound control and explicit CAE observables. ■filecite■turn5file0■L355-L365■ ■filecite■turn5file0■L381-L383■

6. Detailed Mechanism Disclosure

A. Dual-rail ride-down carriage: the seat carriage translates relative to the vehicle structure through a guided path. Each side carries a defined portion of the reaction load and can include an absorber or a redundant structural path.

B. Independent seatback rotation control: the seatback frame is constrained by a separate bilateral path that can include links, dampers, torsional elements or hard stops. This path is intentionally not identical to the energy-stroke path.

C. Multi-state lock: the lock mechanism is disclosed generically to permit cam/pawl, wedge, dog-clutch, pin, ratchet or equivalent constructions. The inventive point is its state coordination with ride-down and rebound, not the existence of an inertial pawl itself.

D. Energy absorber: the replaceable cartridge is characterized by $F(x)$, $F(t)$, hysteresis and temperature response. The design package uses the nominal 180 mm concept stroke, but this remains a design assumption until derived from a vehicle-specific crash pulse and validated energy balance. ■filecite■turn5file0■L365-L370■

E. Occupant trajectory: the system is disclosed as a coupled seat + restraint + occupant problem. The CAE observables include seat travel, seatback rotation, head excursion, pelvis travel/rotation, reaction forces and absorber work.

F. Effective-gravity language: the equation $g_{\text{eff}}(t)=g-a_{\text{seat}}(t)$ is retained as an inertial-frame interpretation only. It must not be described as a local gravitational field generator. ■filecite■turn9file0■L655-L655■

7. Temporal State Machine - Turning "Time" into the Invention

State	Illustrative interval	Mechanical action	Observable evidence
S0 Normal	Before event	Seat adjustment / conventional restraint.	Position, lock state.
S1 Armed/Capture	Early event	Trigger prepares controlled release while primary restraint is available.	Trigger state available.
S2 Ride-down	Primary stroke	Seat translates along dual rails; absorber enters characterized stroke region.	Travel, absorber force.
S3 Rotation control	During ride-down	Distinct path limits seatback rotation.	Back angle, torque, side-link load.
S4 Rebound control	After primary stroke	Directional damping/locking limits reverse motion.	Reverse velocity, damper force.
S5 Secure	Post-event	System reaches end-stop/secure/service condition.	Residual travel, lock state, inspection condition.

The key drafting point is that the exact milliseconds are not the invention by themselves. The invention is the mechanical consequence of a time/event sequence: the seat changes state and therefore changes which force path is active. This makes the disclosure more defensible than claiming a fixed "0-400 ms" recipe.

8. Claim-to-Validation Matrix

Claim limitation	Physical / CAE observable	Validation stage
Controlled longitudinal displacement	Seat travel $x(t)$, velocity, absorber $F-x$	V1 reduced-order -> V2 component -> V4 sled
Separate rotation-control path	Seatback angle $\theta(t)$, link loads, hinge moment	V1 -> V4 -> V5 FE/HBM
Bilateral load paths	S1/S2 reactions, asymmetry, torsional response	V0 CAD -> V1 MBD -> V4/V5
Rebound-control state	Reverse travel/velocity, damping force, secondary excursion	V1 -> V4
Restraint path remains operable	Belt force/timing and anchorage loads	V4/V5
Pelvic trajectory control	Pelvis x/z , rotation, submarining metrics	V4/V5
Absorber characterization	$F-x$, hysteresis, temp sensitivity, cycle repeatability	V2 component test
Structural integrity	Plastic strain, buckling, joint/bolt utilization	V5 FE + physical component tests
Vehicle-level safety	ATD/HBM injury metrics + compliance	V6 full-vehicle

9. Source-Derived Engineering Basis

The supplied THOR-05F material provides a model for professional validation discipline rather than a technology source for the invention. It shows that advanced occupant testing requires extensive instrumentation, defined seating posture, calibration/qualification, repeated checks and post-test inspection. ■filecite■turn2file4■L286-L299■

The supplied PMHS rear-facing frontal study is particularly useful because it shows why a single scalar force target is inadequate: rib fracture distribution depended on seatback properties and combined thoracic loading, while pelvic outward deformation was associated with off-axis rotations. ■filecite■turn5file3■L458-L458■

The 2025 NHTSA/SAE biofidelity work also emphasizes the distinction between sled and full-vehicle conditions and the limited injury conclusions possible from a small sample. ■filecite■turn5file4■L499-L503■

The FMVSS 214 laboratory procedure in the supplied document is treated as a methodological reference for side-impact test discipline, not as a universal frontal/rear seat design limit. The document itself states that an OVSC laboratory test procedure is not itself a regulation or NHTSA interpretation. ■filecite■turn5file2■L378-L378■

10. Rear-Seat Embodiments - 2+1 and 3-Seat

2+1: two outboard T-OCS modules can use dedicated carriage/absorber/load paths; the center seat can use a narrower or shared structural architecture while preserving independent occupant restraint and appropriate egress/packaging.

3-seat: three independently managed modules can share selected cross-members while keeping independent absorption/locking functions. The claim set should not require exactly three seats; seat count is packaging, not the inventive center.

Commercial rationale: the same state-transition logic can be packaged into sedan, SUV, crossover, MPV and autonomous-vehicle interiors. That gives the manufacturer a platform architecture rather than a single seat SKU.

Ruthless caveat: the industrial-design renderings are design-intent visualizations. They do not prove H-point compliance, belt anchorage compliance, occupant clearance, manufacturability, or crash performance.

11. Front-Seat Embodiment

For a front driver/passenger seat, the same T-OCS architecture can be packaged under the seat structure, but vehicle-specific constraints become much tighter: steering-column and pedal clearance, airbag interaction, seat-track hardpoints, wiring, service access, and regulatory requirements must be resolved. The THOR-05F research material illustrates why lower-extremity and thorax loading must be evaluated together rather than reduced to one scalar measure.

■filecite■turn2file4■L294-L299■

The patent should therefore protect the mechanism family rather than a single front/rear geometry. Front and rear packaging are embodiments of the same temporal architecture.

12. Safety / Fail-Safe Disclosure

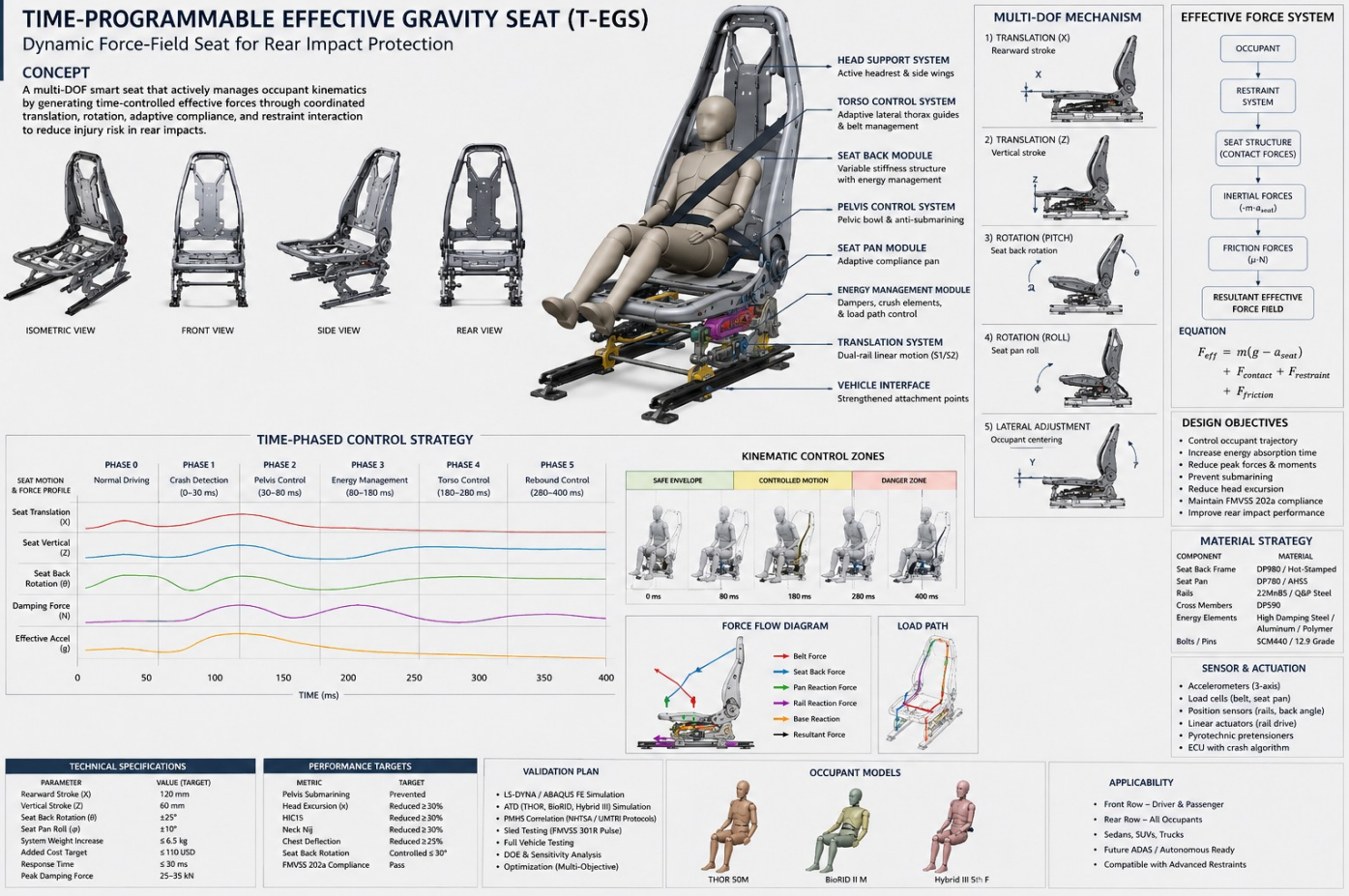
The primary occupant restraint path remains available even if the optional ride-down trigger does not arm. The system should default to a mechanically safe state when trigger, sensor, actuator, or power availability is lost.

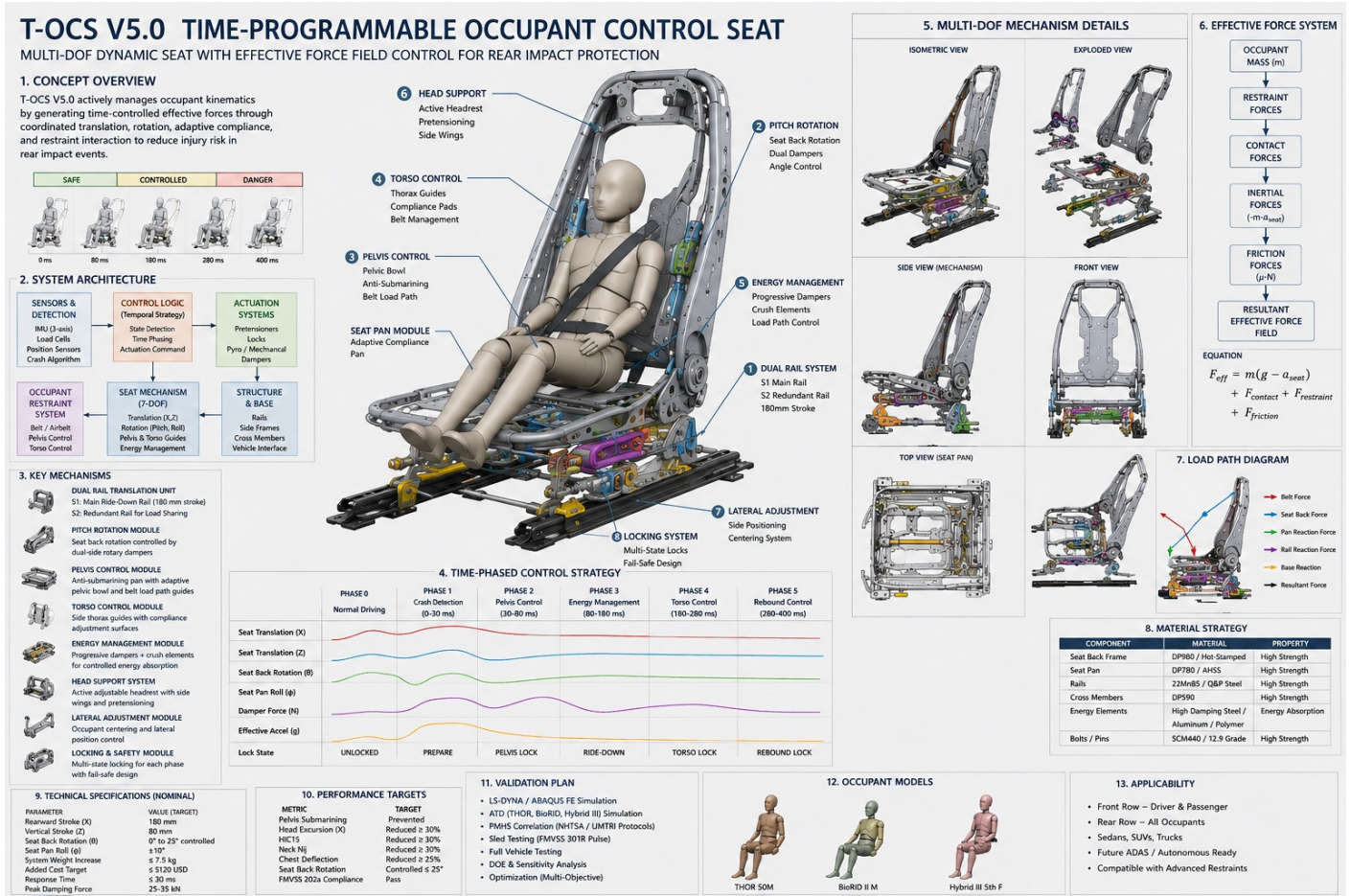
Required design reviews before any human-subject or vehicle-crash use:

- Single-point failure and common-cause failure analysis for the lock/trigger.
- Mechanical stop analysis at both stroke ends.
- Over-travel and jam analysis for both rails.
- Asymmetric-load analysis for one-sided absorber or link degradation.
- Temperature and contamination sensitivity of seals, springs, damping fluid and locks.
- Belt/airbag interaction and anchorage load validation.
- Post-crash inspection/replacement rules for serviceable absorber cartridges.

13. Patent Figure Set - Latest T-OCS Development Boards

The following figures document the evolving design embodiments. They should be redrawn as consistent patent figures before filing, with stable reference numerals and without relying on decorative labels or rendered colors as the only disclosure.







INDUSTRIAL DESIGN – PRODUCTION READY

ISOMETRIC VIEW

SIDE VIEW

FRONT VIEW

REAR VIEW

REAR SEAT CONFIGURATIONS

المقاعد الخلفية

المواد والتشطيبات

- إطار رئيسي: DP980 / QAP Steel
- غطاء على 22MnB5
- عناصر دقيقة: 6061 T6 Aluminum
- عناصر امتصاص: Steel + Polymer
- وسائد: PU Foam + Leather / Fabric
- تشطيب: E-Coat + Powder Coating

مميزات التصميم الصناعي

- تصميم مدمج وخفيف
- تركيب وسادة سهلة
- تشطيبات عالية الجودة
- تكاميل مع أنظمة السيارة
- راحة عالية وتهدئة محسنة
- توافق مع جميع فئات المركبات

الأبعاد الرئيسية (مم)

خيارات الألوان والخامات

أسود

رمادي

بيج

بنّي

جلد طبيعي

جلد صناعي

قماش ناعم

ميكروفايبر

مخرجات CAD – جاهز للتصنيع

النماذج ثلاثية الأبعاد

STEP

IGES

CATIA

SOLIDWORKS

NX

للوصول إلى نموذج لقطعي ثلاثي الأبعاد

الرسومات الهندسية

- 2D DRAWINGS (GD&T)
- ASSEMBLY DRAWING
- BOM – PART LIST

حلول المشكلات الرئيسية

المشكلة	T-OCS V5.0
طيران الجسم للأمام	التحكم الزمني في الانزياح + تثبيت الحوض + إدارة الطاقة
الانزياح تحت الحزام (Submarining)	حوض مطهر + وسائد جانبية + شد مسبق للحزام
إصابة الصدر والظهر	توجيه الذراع الجانبي + تغيير زاوية الظهر + أحزمة متكيفة
إصابة الرأس	دعم رأس نشط + وسادة هوائية جانبية للرأس (اختياري)
الذروة العالية للقوى	عناصر امتصاص طاقة متدرجة + دمبرات قابلة للبرمجة
عدم توافق أحجام الأسماء	تعديل تلقائي للحوض والظهر + حساسات وزن وموضع
الاستدامة والسلامة	مسارات متوازنة + قفل متعدد الحالات + حساسات مراقبة

سلامة أعلى

راحة أعلى

تحكم ذكي

تصميم قابل للتصنيع

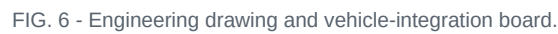
قابل للتكامل

T-OCS V5.0
نظام مقعد ذكي للأمان والتحكم بالحركة

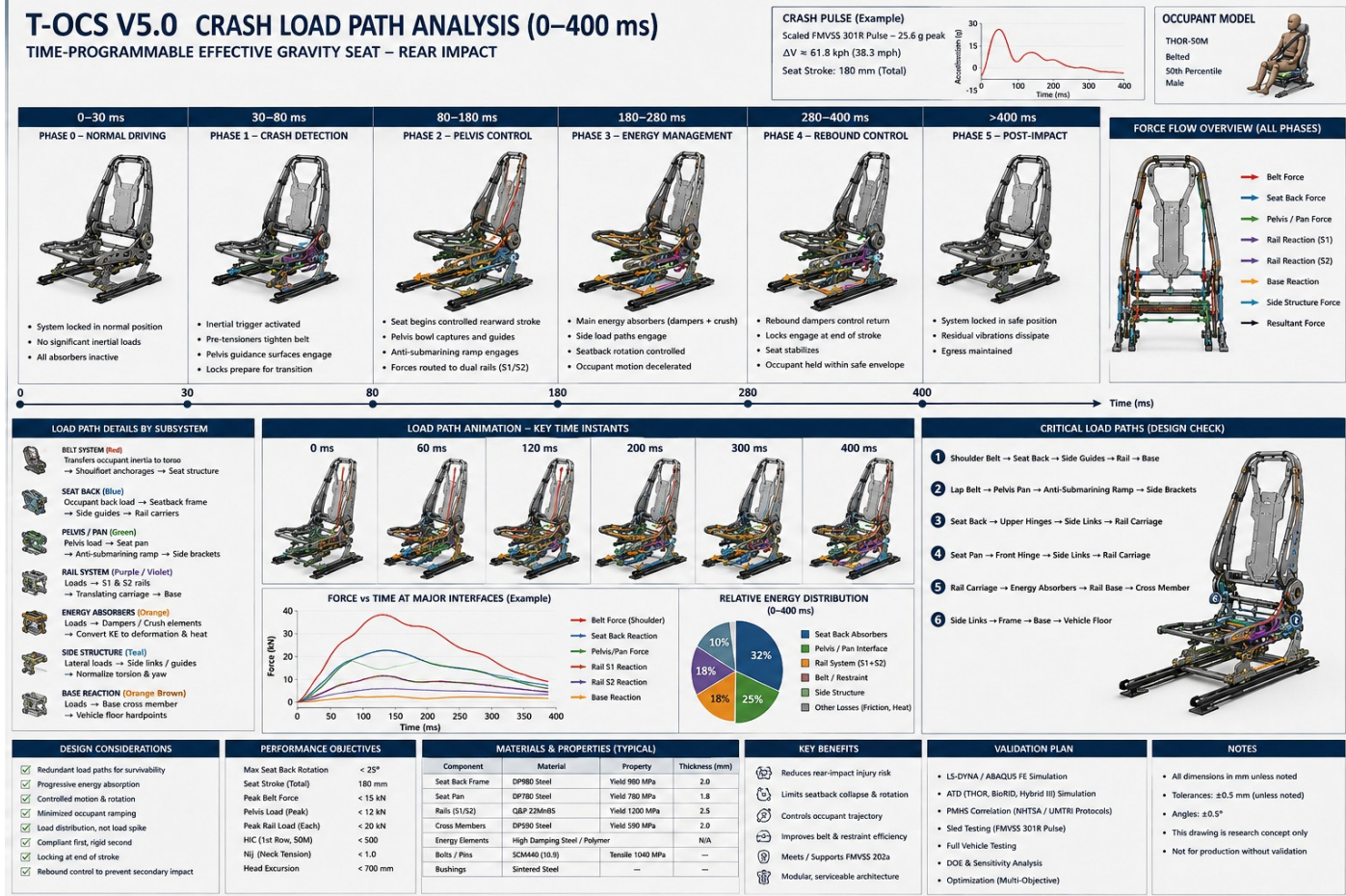
FIG. 3 - Bilingual exploded CAD and industrial-development board.



FIG. 5 - Industrial rear-seat design: 2+1 and 3-seat embodiments.







2. ENERGY ABSORPTION MODULE – EXPLODED

-
- This diagram shows an exploded view of a vehicle chassis. The main components are labeled with blue numbers 1 through 22. The chassis includes a central frame, two side rails, and various mounting brackets and bolts. The components are arranged in a way that shows their relative positions and how they fit together.

All dimensions are in mm.
Scale: NTC

Technical drawing of the Energy Absorber (Type A) showing components and force-stroke characteristics.

Components:

- Cartridge Housing: High strength steel tube
- Guide Rod: Hardened steel
- Piston: Aluminum
- Main Coil Spring
- Hydraulic Damper: Velocity sensitive
- End Cap: CNC machined
- Preload Adjuster: External adjustment
- Mounting Clevis: High strength steel

Dimensions:

- 420 (Extended)
- 240 (Compressed)

FORCE-STROKE CHARACTERISTIC (TYP.)

Graph showing Force (kN) vs. Stroke (mm). The curve shows a non-linear relationship, starting at 0 kN for 0 mm stroke and reaching approximately 20 kN at 200 mm stroke.

Stroke (mm)	Force (kN)
0	0
50	~5
100	~12
150	~17
200	~20

ENERGY ABSORBER PERFORMANCE (Per Side)

Peak Force @ 25.6 g	18 – 22 kN
Stroke	180 mm
Shock Absorption Capacity	2.0 – 2.8 kJ

- Dual-rail ride-down with progressive energy absorption
- Passive inertial trigger with mechanical fail-safe
- Multi-state locking (Drive / Prepare / Ride-down / Locked)
- Controlled rebound via dedicated dampers
- Serviceable modular cartridges
- Built for crash loads up to 25.6 g (scalable)

Ride-down stroke	180 mm
Total travel	210 mm
Seat back rotate (max)	- 50°
Pitch range	-10° to +25°
Roll range	±8°

LOCK STATES & FUNCTION				
STATE 0 DRIVE (UNLOCKED)	STATE 1 PREPARE (ARMED)	STATE 2 RIDE-DOWN (LOCK POSITION)	STATE 3 END-STOP (SECURE)	
Free motion All DOF active Conductor mode	Inertial weight Armed Load engaged	Red engaged Guides locked Pawl engaged with ratchet	End-stop engaged Full lock Crash lock secured	

520
480
180 (Stroke)

- 11/22A Outer Rail DP780 / Hot Stamped
- 11/22B Inner Rail DP780
- 11/22C Ball Cage Steel
- 11/22D End Stop Steel
- 11/22E Lubrication Port Grease Fitting

**DETAIL C
FRONT HINGE**

M12 BOLT - CLASS 10.9
TORQUE: 95 Nm
NUTTING: 1012-1210/0

**DETAIL D
DUAL RAIL MOUNT**

M10 BOLT - CLASS 8.8
TORQUE: 55 Nm
WELD NUT - 4010

**DETAIL E
ABSORBER MOUNT**

M12 BOLT - CLASS 10.9
TORQUE: 95 Nm
NUTTING: 1012-1210/0

JOINT TYPE	WELD TYPE	SIZE	LENGTH	NOTES
Frame to Rail	MIG	6 mm	Continuous	Both Sides
Side Bracket	MIG	5 mm	60 mm	Stitch-50
Cross Member	MIG	6 mm	Continuous	Both Sides
Seat Pan	MIG	4 mm	Continuous	Stitch OK
Mounting Tab	MIG	5 mm	40 mm	Stitch-40

WELD CONSUMABLE: ER70S-6
SURFACE PREP: Grind & Clean before welding

Pitch Hitch Adaptor
Adjustment Range: -10° to +25°

- 1 Pitch Side Plate (L/R) (mm)
- 2 Pivot Shaft (1/8 mm - 10.9)
- 3 Sector Gear
- 4 Lock Pin Hardened Steel
- 5 Gas Spring / Damper

PITCH POSITIONS

- 10°
- 0°
- +25°

Linear Dimension (0-300)	±0.5 mm
Linear Dimension (300-1000)	±1.0 mm
Hole Diameter	H11
Shaft Diameter	h11
Angle	±0.5°
Parallelism	0.5 mm / 300 mm
Perpendicularity	0.5 mm / 300 mm

- 1 Head Support → Upper Frame → Side Guides → Seat Pan
- 2 Seat Back → Side Links → Side Absorbers → Rails
- 3 Pelvis → Seat Pan → Rails → Energy Absorbers → Vehicle
- 4 Seat Pan → Crush Elements → Side Brackets → Rails
- 5 Restraint Loads → Pelvis/Torso Guides → Frame
- 6 Rails → Vehicle Floor Anchors (4 Points)

Top view of the robot chassis showing dimensions: 400mm width, 200mm height, and 380mm wheelbase.

- This drawing package is a research prototype.
- All dimensions are nominal.
- Do not use for production without validation.
- Validate with FE, sled test, and full vehicle test.
- Comply with FMVSS 202a, 207, 301, and other applicable regulations.
- All components must be inspected and tested per validation plan.

REV.	DATE	DESCRIPTION	BY
A	2025-05-10	Initial Concept Release	ENG
B	2025-05-20	Added Locking Module Details	ENG
C	2025-05-28	Added Sections & Tolerances	ENG
D	2025-06-01	Final Research Release	ENG

T-OCS V5.0 – TIME-PROGRAMMABLE EFFECTIVE GRAVITY SEAT – RESEARCH PROTOTYPE – NOT CERTIFIED FOR PRODUCTION

FIG. 9 - Detailed energy absorber module / valve / cartridge package.

14. Prior-Art Source Map - Google Patents Screening

Reference	Relevant disclosure	V6 risk / treatment
US20200231073A1 / US11007912B2	Energy-attenuating seat; track-based energy attenuation; independent energy absorber + pivot.	Highly avoidable (generic energy absorber + pivot).
EP3911536A1 / B1	Related energy-attenuating seat family with rotary attenuation.	High. Same caution.
WO2021197809A1 / US12151595B2	Seat/rail pair + energy absorber + locking device; crash-triggered release/high/avoidable removal.	High/avoidable (generic release/absorber claims).
US4634182A	Inertial recliner lock using pendulum/pawl/cam.	High for generic inertial lock.
US6050635A	Anti-submarining seat-pan structure.	High for generic anti-submarining feature.
US5125598A	Pivoting energy-attenuating seat with crash-induced seat-pan motion.	Medium-high for generic pivoting attenuation.

Google Patents screening is a prior-art search aid, not a legal novelty or infringement opinion. Google itself notes that legal-status information is not a legal conclusion. [■cite■turn992619search0■turn464299search0■](#)

15. V5 -> V6 Claim-Strategy Changes

V5 tendency	V6 correction	Why
"Effective gravity" as novelty hook	Removed from independent-claim center.	Prevents attack on a physically misleading framing.
Broad rail + absorber + lock combination	Narrowed to specific relationships and state transitions.	Known blocks already disclosed.
Generic inertial lock	Dependent embodiment only.	Classic prior art exists.
Seat rotation and translation treated together	Explicitly separated into two mechanical paths.	Creates a clearer architectural distinction.
Rebound mentioned as a phase	Rebound tied to a defined travel/kinematic transition and direction.	Treats rebound as a claim limitation.
2+1 / 3-seat as novelty	Packaging embodiments only.	Seat count is unlikely to carry novelty.
Illustrative performance numbers	Tagged as targets or source-specific values.	Avoids false performance claims.

16. Pre-Filing Gate - Do Not Skip

- Confirm the true earliest conception date and preserve dated CAD files, notebooks, sketches and revisions.
- Identify every public disclosure already made and obtain patent counsel advice on priority, foreign rights and any applicable grace-period rules.
- Run a professional novelty and inventive-step search beyond the preliminary Google Patents screen.
- Create a claim chart mapping each V6 limitation against each relevant prior-art document.
- Convert visual boards into consistent patent drawings with stable reference numerals.
- Replace every nominal CAD dimension with a vehicle-package value, measured prototype value, or explicit assumption.
- Perform V0 CAD integrity and interference review.
- Perform V1 multibody dynamic screening.
- Characterize V2 absorber F-x/hysteresis/temperature/cycling.
- Bench-test V3 lock/trigger including fault and unintended-release cases.
- Run V4 sled tests with appropriate ATDs and high-speed video.
- Correlate V5 FE/HBM models to physical data before making injury-reduction claims.
- Use full-vehicle testing for V6 vehicle-level compliance and safety evidence.

Final ruthless verdict: V6 is a materially stronger patent-preparation package than V5 because its center of gravity has shifted from an attractive concept to a specific, testable architecture. However, there is still no honest basis to promise patent grant, freedom to operate, or crash-safety performance before counsel review and validation.

17. References Used in V6

Uploaded engineering sources:

1. 15874-SAE GIM 2023 THOR05F for distrib - Evaluation of THOR-05F in a Full Vehicle Crash Series. ■filecite■turn2file4■
2. 16616-Tesny_SAE 2025 GI Prelimin Biofidelity Asses_2 - A Preliminary Biofidelity Assessment of THOR and BioRID in a Moderate-Speed Rear Impact Sled Test. ■filecite■turn5file4■
3. 16616-Greb_SAE 2025 Seat Assess In Standard Rear I - Seat Assessment in Standardized Rear Impact Virtual Testing. ■filecite■turn5file5■
4. Thoracic and pelvic responses and injuries to PMHS in rear-facing seat configurations in high-speed frontal impacts. ■filecite■turn5file3■
5. TP-214D-09 - Laboratory Test Procedure for FMVSS No. 214, Dynamic Side Impact Protection. ■filecite■turn5file2■
6. T-OCS V5.0 Engineering Report and Design Parameters. ■filecite■turn5file0■turn5file1■

Google Patents prior-art screening:

- US20200231073A1 - Energy attenuating vehicle seat assembly. ■cite■turn992619search0■
- EP3911536A1 - An energy attenuating vehicle seat assembly. ■cite■turn464299search1■
- WO2021197809A1 - Seat rail pair for longitudinal adjustment of a vehicle seat and energy absorber and locking device. ■cite■turn464299search0■
- US4634182A - Seatback recliner mechanism and inertia operated lock. ■cite■turn992619search1■
- US6050635A - Seat pan of an automobile vehicle seat including an anti-sub-marining device. ■cite■turn992619search2■
- US5125598A - Pivoting energy attenuating seat. ■cite■turn464299search2■

Document status: **Patent-preparation technical draft / invention disclosure, not legal advice and not a filed application.**